SALES DATA CLEANING AND PREPROCESSING USING PYTHON

Project Report

Submitted by:

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Submitted To:
DecodeLabs

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Declaration

I certify that the project report titled "Data Cleaning and Preprocessing for Sales Data Analysis using Python" was written by me as a requirement for the Data Analytics Internship Program at DecodeLabs.

The content included in this report is entirely original except where cited.

Signature: ___Thamindu Kavinda___

Name: Thamindu Kavinda

Date: ______2026/05/19_______

Acknowledgement

I am honored to express my sincere gratitude to DecodeLabs for giving me the opportunity to complete this Data Analytics Internship Project.

Thank you to all the mentors and advisors who guided me during this project.

This project gave me hands-on experience in data cleaning and preprocessing using Python and Pandas.

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1. Introduction

The aim of the current project is to prepare the dataset for analysis by performing various data cleaning processes. Data cleaning is an essential part of the data analysis process as raw data usually has many issues such as missing values, duplicates and other inconsistencies.

The main goal of the current project is to prepare the data before further data analysis is performed. A number of data preprocessing processes are applied.

2. Objectives

- To load and inspect the dataset
- To identify missing values
- To detect duplicate records
- To clean inconsistent data
- To standardize date formats
- To create a cleaned dataset for analysis

3. Technologies Used

- Python
- Pandas
- Google Colab
- Microsoft Excel

4. Dataset Description

The dataset used in the project provided data on sales transactions. The dataset includes details such as order ID, date, coupon codes, and customer details.

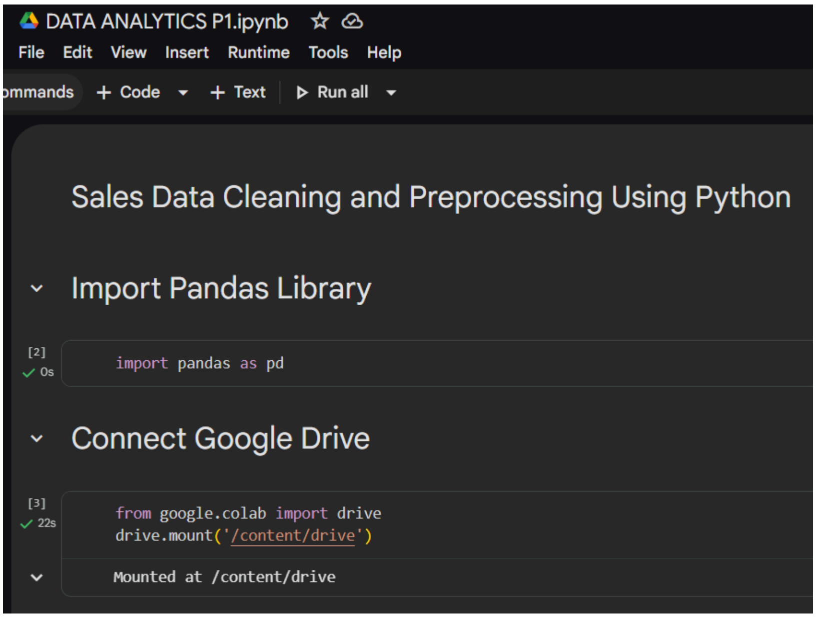
The above dataset was used for data cleaning and preprocessing.

5. Methodology

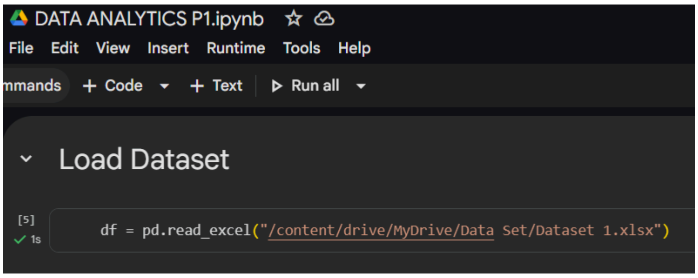
- Importing Libraries
- Loading Dataset
- Dataset Inspection
- Missing Value Detection
- Duplicate Detection
- Removing Duplicates
- Date Conversion
- Filling Missing Values
- Exporting Cleaned Dataset

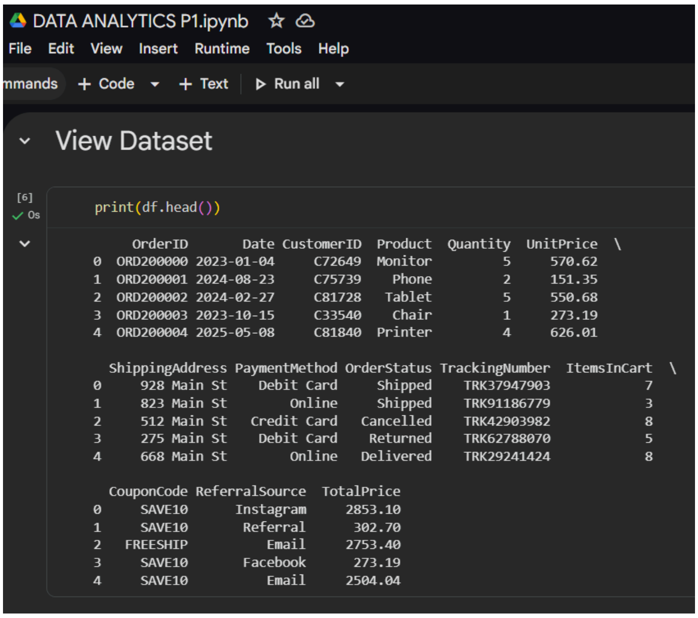
6. Project Screenshot

- Import Library and Connect google drive

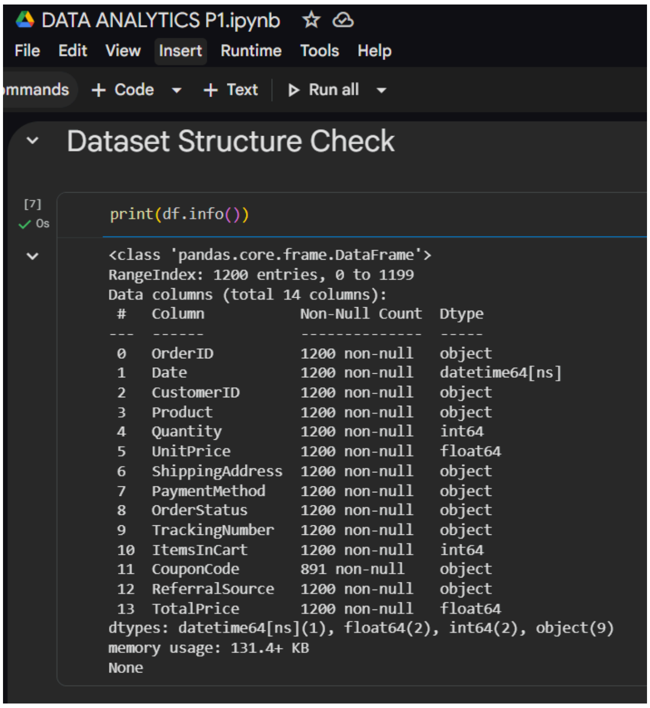


- Loading the Dataset and Dataset Inspection

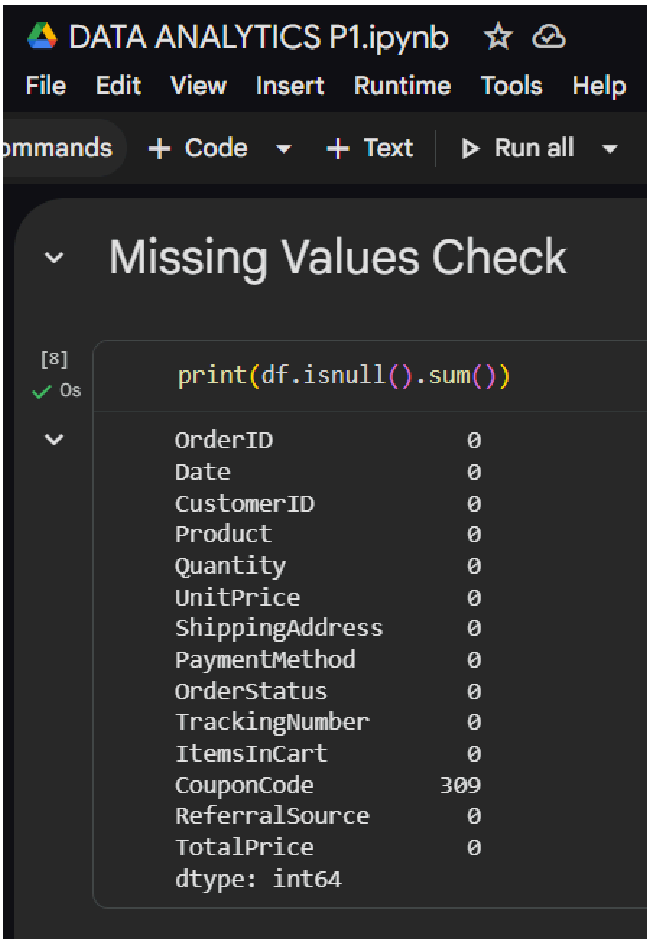




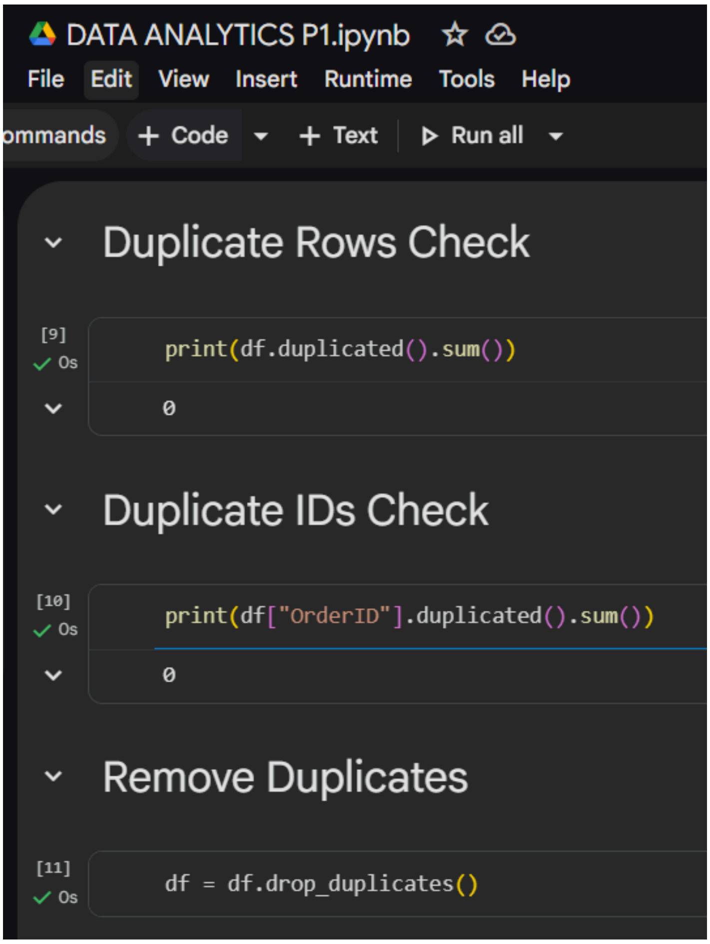
- Checking Dataset Information



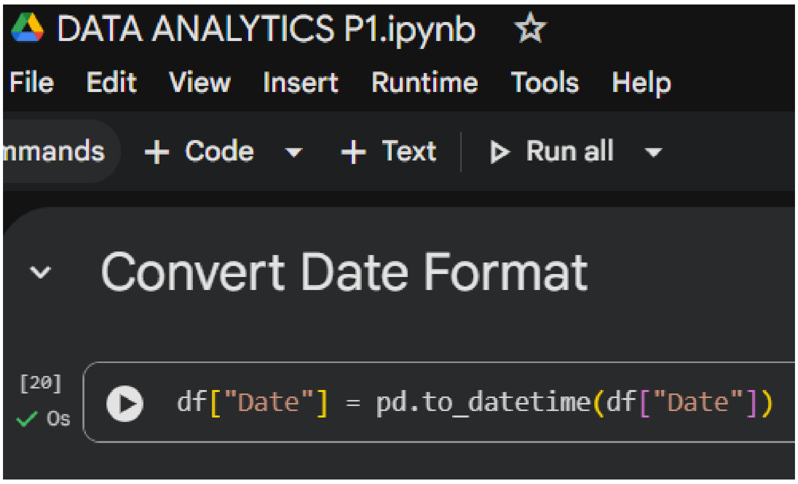
- Missing Value Detection

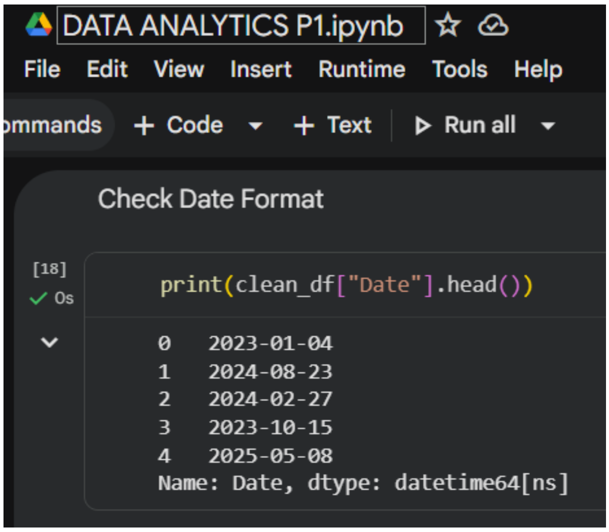


- Duplicate Record Detection and Removing Duplicate Records

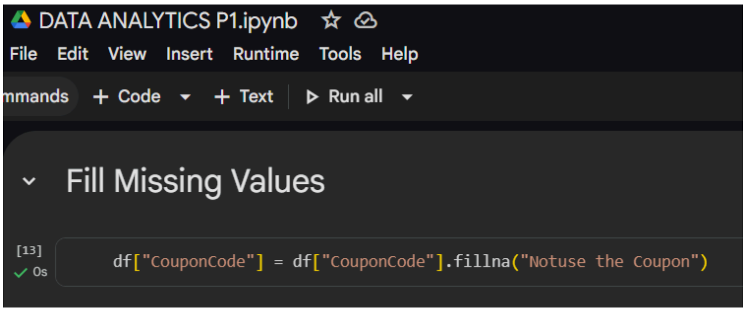


- Date Format Conversion

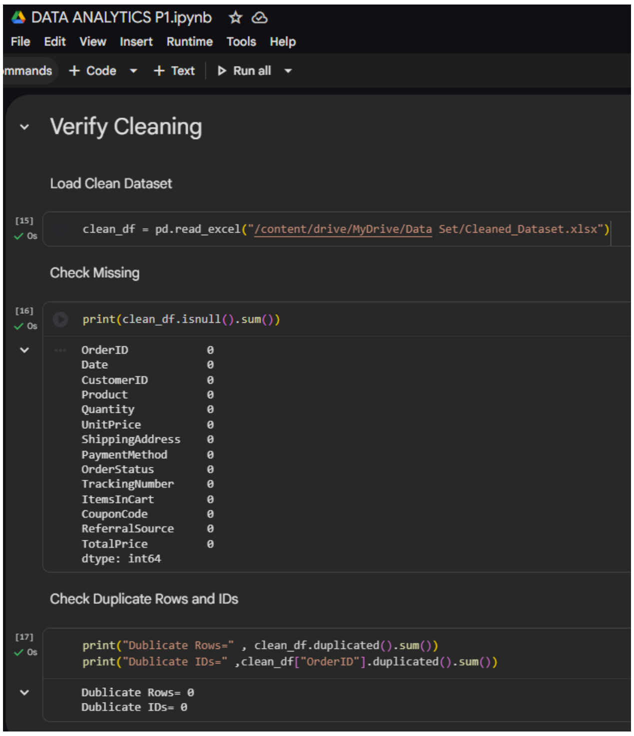




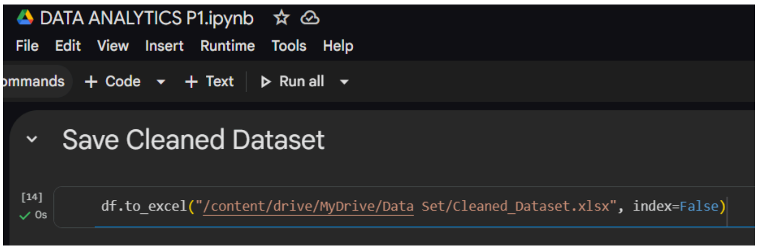
- Handling Missing Values

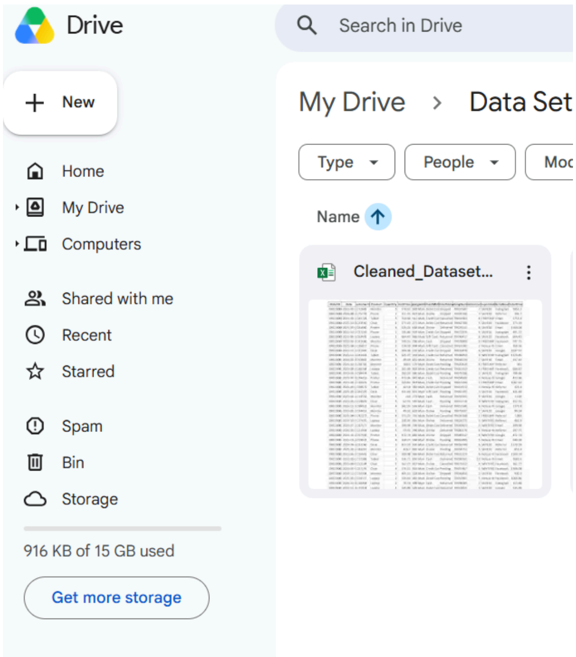


- Verify Dataset

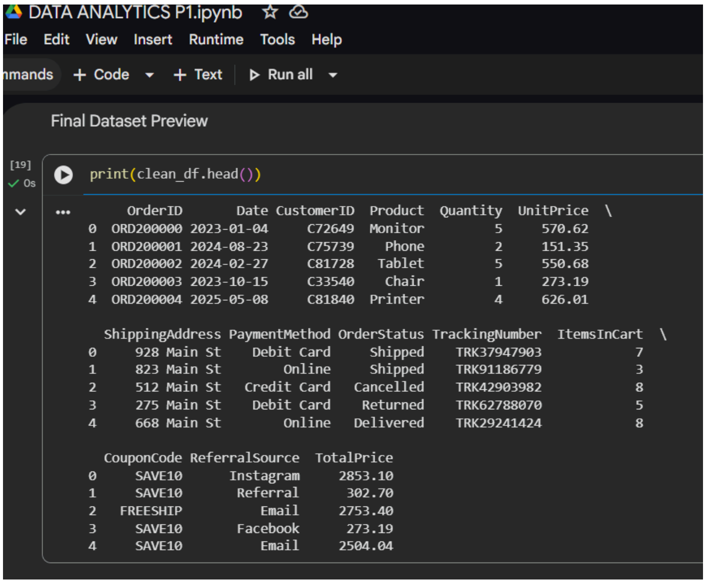


- Exporting the Cleaned Dataset





- Final Cleaned Dataset



DATA ANALYTICS P1.ipynb

File Edit View Insert Runtime Tools Help

Final Dataset Preview

```
print(clean_df.head())
```

```
     OrderID       Date CustomerID  Product  Quantity  UnitPrice  \
0  ORD200000 2023-01-04     C72649  Monitor         5     570.62
1  ORD200001 2024-08-23     C75739    Phone         2     151.35
2  ORD200002 2024-02-27     C81728   Tablet         5     550.68
3  ORD200003 2023-10-15     C33540    Chair         1     273.19
4  ORD200004 2025-05-08     C81840  Printer         4     626.01

  ShippingAddress PaymentMethod OrderStatus TrackingNumber  ItemsInCart  \
0     928 Main St    Debit Card     Shipped    TRK37947903            7
1     823 Main St        Online     Shipped    TRK91186779            3
2     512 Main St   Credit Card   Cancelled    TRK42903982            8
3     275 Main St    Debit Card    Returned    TRK62788070            5
4     668 Main St        Online   Delivered    TRK29241424            8

  CouponCode ReferralSource  TotalPrice
0     SAVE10      Instagram     2853.10
1     SAVE10       Referral      302.70
2   FREESHIP          Email     2753.40
3     SAVE10       Facebook      273.19
4     SAVE10          Email     2504.04
```

7. Conclusion

The project emphasized the importance of data cleaning in the data analysis process. Various methods of data preprocessing were used using the programming language Python and its library Pandas.